Smart Streetlights for Smart Connectivity: A Solution to Bloemfontein's Urban Digital Divide

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DECLARATION WITH REGARD TO INDEPENDENT WORK

I, Sizwe Marole identity number 0108175401083 and student number 222008531, hereby declare that this research project submitted to the Central University of Technology, Free State, for the degree Advanced Diploma in INFORMATION TECHNOLOGY, is my own independent work. I further declare that it complies with the Code of Academic Integrity, as well as other relevant policies, procedures, rules, and regulations of the Central University of Technology. In addition, I confirm that it has not, in part or as a whole, been submitted before to any institution by myself of any other person in fulfilment (or partial fulfilment) of the requirements for the attainment of any qualification.

SIGNATURE OF STUDENT ----------------------------- **DATE:15/05/2025**

ABSTRACT

In the digital era, affordable access to the internet is important for social inclusion, education, and economic participation. However, in South African cities like Bloemfontein, high mobile data costs and limited broadband infrastructure have deepened the urban digital divide, excessively affecting low-income communities. This study explores the integration of Wi-Fi technology into existing streetlight infrastructure as a cost-effective and scalable solution to improve public internet access. By leveraging smart streetlights as digital access points, municipalities can extend connectivity to underserved public spaces such as parks, taxi ranks, and informal settlements. Implementing a mixed-methods research design, the study combines quantitative data, such as Wi-Fi coverage mapping and household surveys, with qualitative insights from stakeholder interviews and community focus groups. The research aims to evaluate the possibility, benefits, and social impact of streetlight-based Wi-Fi in Bloemfontein, providing policy recommendations for development in digital inclusion. Findings will contribute to ongoing smart city initiatives and offer practical strategies for addressing infrastructure challenges in digitally underserved urban environments.

LIST OF ABBREVIATIONS

4IR	Fourth Industrial Revolution
ISP	Internet Service Provider
Wi-Fi	Wireless Fidelity
ICT	Information and Communication Technology
FTTH	Fibre-to-the-Home
LAN	Local Area Network
IEEE	Institute of Electrical and Electronics Engineers
SA Connect	South Africa Connect (National Broadband Policy)
GIS	Geographic Information Systems
SPSS	Statistical Package for the Social Sciences
Stats SA	Statistics South Africa
IoT	Internet of Things
UN	United Nations

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Chapter 1: Introduction

1.1 Background

In today's digital age, internet connectivity is essential for economic growth, education, and social inclusion. According to UN estimates, over 68% of the global population will be living in urban areas by 2050, and this urbanization brings distinct challenges. Urban centers are under increasing pressure to provide inclusive and affordable digital infrastructure (Gehlot et al., 2021). Traditional broadband expansion is often costly and slow, particularly in low-income areas.

One emerging solution to address this gap is the integration of Wi-Fi technology into existing streetlight infrastructure. This innovation allows streetlights to function beyond their conventional role by serving as digital access points, extending internet connectivity to public spaces such as parks, streets, and transportation hubs. By leveraging existing urban infrastructure, municipalities can expand internet coverage in a cost-effective and scalable manner, supporting the vision of smart, connected urban areas.

This study aims to evaluate the potential of Wi-Fi-enabled smart streetlights to address the digital divide in urban areas, particularly in low-income neighborhoods. Using a mixed-methods approach, the research will assess the feasibility, benefits, and challenges of integrating Wi-Fi routers into streetlight infrastructure and propose a scalable model for municipal adoption to enhance digital inclusion.

1.2 Problem Statement

Despite the ongoing advancements brought by the Fourth Industrial Revolution (4IR), traditional broadband infrastructure such as fibre-optic networks and standalone Wi-Fi hotspots remains costly and difficult to scale, particularly in underserved urban areas. In South Africa, where mobile devices are the primary means of internet access, the inflated cost of mobile data continues to restrict connectivity. According to (Matone,2024), 1GB of data can exceed 2% of the average monthly income, making internet access unaffordable for many.

This issue is particularly concerning in cities like Bloemfontein, where rapid urbanization is placing additional pressure on already strained infrastructure. Many communities lack affordable, reliable internet access, further deepening the digital divide and limiting access to education, employment, and socioeconomic opportunities.

One emerging and scalable solution is the integration of public Wi-Fi technology into smart streetlight infrastructure. By leveraging existing municipal assets such as streetlights, municipalities like Bloemfontein can cost-effectively expand internet access to public spaces, including parks, taxi ranks, and densely populated areas. This study aims to explore the potential of Wi-Fi-enabled smart streetlights as a sustainable approach to bridging the digital divide in Bloemfontein's urban communities.

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction to Digital Connectivity and the Digital Divide

In the digital age, unbiased internet access is important to economic development, education, and social engagement (Mlaba Khanyi, 2021). As populations grow, innovative solutions are essential to bridge the digital divide. Integrating Wireless Fidelity (Wi-Fi) technology into streetlight infrastructure presents a promising approach to enhancing public connectivity in South Africa's urban areas, including Bloemfontein. While specific data on Bloemfontein's internet connectivity challenges is limited, regional trends in South Africa suggest that urban areas may face similar issues. For instance, the Independent Communications Authority of South Africa (ICASA) reports that fibre-to-the-home (FTTH) penetration has surpassed 20% in urban areas, indicating a significant portion of the population still lacks high-speed internet access (Independent Communications Authority of South Africa, 2023).

2.2. The Role and Importance of Wi-Fi Technology

Wi-Fi has become a crucial technology in today's interconnected society, reforming how we access and interact with people, businesses, and information. This wireless communication standard enables devices such as smartphones, laptops, and tablets to connect to the internet or local area networks (LAN) without physical connections, using radio waves to carry data between devices and access points. The IEEE 802.11 standard governs Wi-Fi and, due to its widespread use in homes, offices, cafes, and public spaces, has significantly transformed our daily lives (Greedts Christopher, 2016).

2.3. Internet Accessibility Challenges in South Africa

The importance of Wi-Fi in the current generation cannot be overstated. It offers a portal to communication, education, and vital information by enabling seamless connectivity, especially as the rapid development of digital technologies makes continuous internet access essential (Pahlavan and Krishnamurthy, 2021). However, the digital divide remains a significant challenge in South Africa, with many people unable to access the internet due to inflated costs and poor infrastructure. In response, the government launched the SA Connect broadband initiative, which aims to provide 80% of households with high-speed, reliable, and affordable internet by 2026. In addition to expanding connectivity in rural areas such as Chitwa, Bhonga, Lugelweni, and Dutyini in the Eastern Cape, the project has connected several government services, including schools and healthcare facilities (Ditlhake Matone, 2024). In Bloemfontein, where connectivity is increasingly important for educational and economic opportunities, this initiative could make a significant difference. SA Connect has already created over 4,500 jobs, reduced data costs, and generated employment opportunities, particularly in the ICT sector. South Africa is striving to close the digital divide through many initiatives, promoting increased socioeconomic development and inclusivity (Masia Laticia, 2024).

2.4. Government Initiatives: SA Connect and Digital Inclusion

As the need for affordable and reliable internet access becomes increasingly critical, cities worldwide are exploring advanced solutions to develop public connectivity. Over the past two decades, South Africa has undergone significant technological and political transformation. While internet connectivity has drastically improved (Donati Dante, 2023), integrating Wi-Fi into streetlight infrastructure provides an opportunity to utilize existing urban resources to promote digital inclusion. However, several factors need to be considered to fully understand the feasibility and potential impact of such solutions (Agramelal et al., 2023).

2.5. Smart Solutions: Integrating Wi-Fi into Streetlight Infrastructure

This research proposes a solution that leverages existing streetlights as Wi-Fi hotspots to improve coverage across Bloemfontein urban areas without the need for significant new infrastructure. Integrating Wi-Fi transmitters into streetlights can extend Internet coverage across larger areas, significantly reducing infrastructure costs (Orejon-Sanchez, Andres-Diaz, and Gago-Calderon, 2021). This approach reduces infrastructure costs and bridges the digital divide by providing Wi-Fi access to underserved communities (Greedts Christopher, 2016).

2.6. Benefits of Streetlight-Based Wi-Fi Networks

Integrating Wi-Fi into streetlight infrastructure has been explored in several studies as a promising solution to bridge the digital divide in underserved urban areas. Streetlights, which are already widespread and integral to city infrastructure, present a cost-effective opportunity for expanding Internet access. By utilizing these existing structures, we can provide connectivity without the need for expensive new infrastructure. This approach not only supports the practical needs of urban connectivity but also aligns with the principles of smart city development, which leverage information and communication technologies (ICT) to enhance urban services like transportation, energy, and public safety, all while promoting digital inclusion (Biundini et al., 2024).

2.7. Urban Connectivity and Smart City Development

Wi-Fi technology in urban areas can promote more fair access to digital resources. As mobile data prices continue to rise, many low-income communities remain excluded from the benefits of the internet. Wi-Fi networks deployed through streetlights offer a lower-cost alternative, enabling greater accessibility for marginalized communities. In regions where internet penetration is low, particularly in areas with high poverty rates, such networks could significantly improve opportunities for education, employment, and social engagement.

2.8. Socioeconomic Impact and Community Empowerment

The goal of this project is to ensure that individuals, especially those in underserved areas, have the same access to the internet as those in more connected regions. In today's digital age, internet access is necessary for job opportunities, education, and participation in the global economy (Dordley Lucinda, 2018). Wi-Fi subscriptions are often more affordable than mobile data plans, making this solution a viable option for bridging the gap. By expanding internet access through this initiative, we can equip individuals with the tools they need to learn, grow, and succeed.

2.9. Implementation Strategy and Coverage Potential

To enhance the impact of this project, it is proposed that Wi-Fi routers be implemented on all streetlights in urban areas, depending on coverage needs (Oza, Hinsu, and Goradiya, 2021). This will ensure that people have access to affordable, unlimited Wi-Fi that supports their daily needs and enables them to stay connected wherever they go. This approach presents the potential to having Internet access everywhere. This initiative could not only improve digital access for all but also position South Africa as a leader in providing affordable public Wi-Fi, accessible to everyone at an affordable price (du Toit and Stimie, 2023).

2.10. Positioning South Africa as a Leader in Public Wi-Fi Access

Beyond bridging the digital divide, this project will stimulate the economy by providing reliable and accessible Wi-Fi, contributing to economic growth and innovation. The widespread availability of Wi-Fi will also facilitate progress in the Fourth Industrial Revolution (4IR) and beyond, as connectivity is an essential tool for all in the modern world. By integrating Wi-Fi into streetlight infrastructure, we are not only enhancing connectivity but also ensuring that every citizen in Bloemfontein has equal access to education, employment, and participation in the digital economy, fostering greater inclusion and a more connected society (Biundini et al., 2024).

2.11. Contribution to Economic Growth

This research aims to explore the potential of using streetlights as Wi-Fi access points to bridge the digital divide in urban South Africa, with a particular focus on Bloemfontein. By leveraging existing infrastructure, municipalities can expand internet access in a cost-effective and scalable manner, promoting greater digital inclusion. Wi-Fi-enabled streetlights can provide reliable internet access to underserved communities, supporting educational, economic, and social opportunities for those currently excluded from the digital world (Independent Communications Authority of South Africa, 2023).

2.12. Conclusion: Towards a More Inclusive Digital Society

Through this study, we aim to provide insights and recommendations for policymakers and urban planners on how to implement streetlight-based Wi-Fi networks in South African cities. Successfully integrating Wi-Fi into streetlights could pave the way for a more connected and inclusive future, where access to the digital economy is no longer a privilege for the few but a right for all.

Chapter 3: Research Framework

3.1 Research Aims:

This research will address several key questions:

• What are the benefits of using streetlights as Wi-Fi access points in low-income underserved areas?

• How can Wi-Fi-enabled smart streetlights contribute to digital inclusion and the development of smart city infrastructure in Bloemfontein?

• How does the lack of reliable internet access impact social and economic opportunities in Bloemfontein?

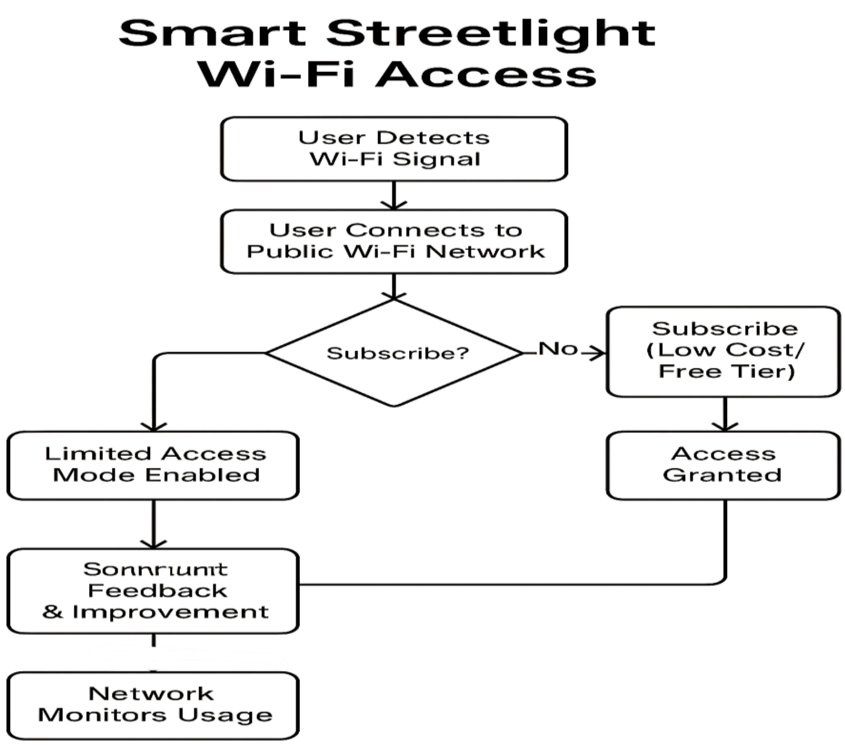
3.2 Objectives:

• To assess the extent of the digital divide in Bloemfontein.

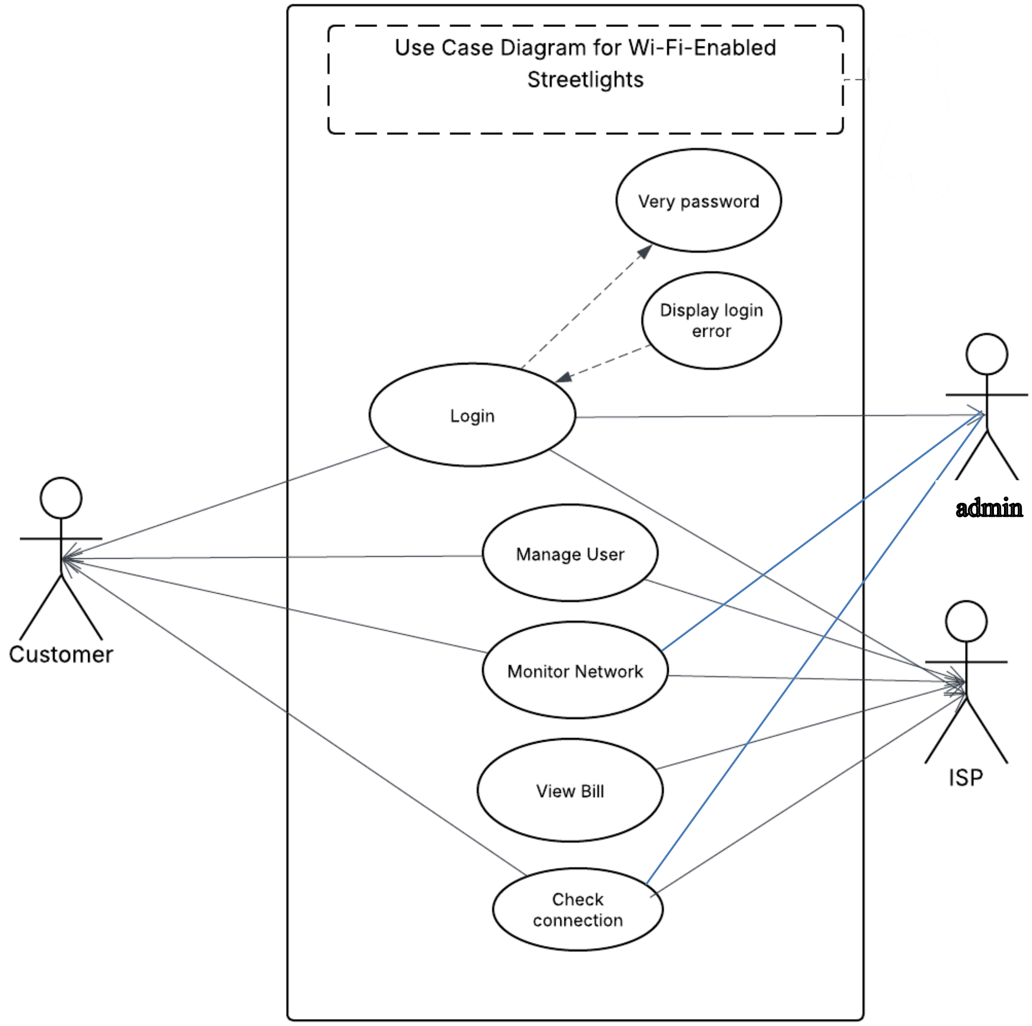
• To evaluate the scalability and cost-effectiveness of streetlight-based Wi-Fi.

• To explore how improved internet access can promote digital inclusion and socioeconomic development.

3.3 Flow chart



3.4 Use Case



3.5 Overall Aim

The overall aim of this research is to investigate the possibility and impact of Wi-Fi-enabled smart streetlights as a scalable and cost-effective solution to bridge the digital divide and support smart city development in Bloemfontein.

CHAPTER 4: METHODOLOGY

4.1. Research Design Overview

This study adopts a mixed-methods research design, combining both quantitative and qualitative approaches to explore the current landscape of public Wi-Fi availability and the integration of smart streetlight infrastructure within Bloemfontein's urban areas. The rationale for this approach lies in the need to not only analyse measurable patterns but also to gain insight into life, individual experiences surrounding internet access in the city.

By combining both numeric data and personal stories, the research ensures a complete perspective that acknowledges the significance of human context alongside technological data. This methodology strengthens the total strength of the findings through triangulation, allowing for verification across multiple data sources and perspectives. The mixed-methods framework enhances the depth and credibility of the research, making it suitable for addressing complex issues at the intersection of technology, society, and infrastructure.

4.2. Rationale for Mixed-Methods Approach

The mixed-methods approach is crucial for understanding the interplay between technological infrastructure Wi-Fi availability and societal factors such as digital exclusion and community needs. Quantitative data provides measurable evidence of Wi-Fi coverage, user patterns, and demographics, while qualitative data allows for deeper understanding through personal experiences and stakeholder insights. This combination provides a strong framework for examining both the technical and social dimensions of the research topic.

4.3. Quantitative Research Component

4.3.1 Wi-Fi Coverage and Digital Exclusion Mapping

The quantitative component focuses on understanding the extent of Wi-Fi coverage and the patterns of digital exclusion in selected urban areas. This includes mapping public Wi-Fi hotspots using data obtained from telecommunications providers and publicly available government reports, notably those published by Statistics South Africa (Stats SA).

A randomized sampling strategy will guide the selection of urban zones with varying degrees of connectivity. Within these zones, the study will assess the number of active users, the spatial distribution of hotspots, and the influence of demographic factors, particularly income and internet accessibility.

4.3.2 Survey Questionnaire and Sampling Strategy

To support this analysis, structured survey questionnaires will be administered to households. These surveys will gather information on internet usage habits, income brackets, and barriers to accessing affordable connectivity. A stratified random sampling technique will be applied to ensure representation across income levels and household types. This data will help uncover which socioeconomic groups are most affected by the inflated cost of data plans or lack of infrastructure.

4.4. Data Analysis and Tools

Quantitative data will be analysed using statistical software such as SPSS or Microsoft Excel, enabling the generation of charts, graphs, and cross-tabulations. In addition, Geographic Information Systems (GIS) will be used to visualize hotspot coverage, identify connectivity gaps, and map high-demand zones in underserved communities.

4.5. Qualitative Research Component

4.5.1. Stakeholder Interviews

To understand the social and infrastructural implications of smart streetlight-enabled Wi-Fi, the qualitative phase involves engaging with stakeholders and community members through semi-structured interviews and focus group discussions. Key stakeholders will include urban planners, ICT policy advisors, municipal infrastructure officers, and community leaders. These interviews aim to explore the possibility, potential benefits, and implementation challenges of integrating Wi-Fi into public lighting infrastructure. Participants will be purposively selected based on their experience and relevance to the research focus.

4.5.2 Focus Group Discussions

Focus groups will be held with residents from selected communities, ensuring diverse representations such as students, informal traders, older adults, youth, and small business owners. These discussions will explore everyday experiences with internet access, perceptions of smart infrastructure, and community priorities.

4.5.3 Case Studies

To deepen the qualitative analysis, case studies will be conducted in areas where smart streetlight Wi-Fi has been successfully introduced. These will serve as practical examples, highlighting the best practices and lessons learned.

4.6. Data Analysis for Qualitative Research

Collected qualitative data will undergo thematic analysis, where key themes such as accessibility, infrastructure challenges, digital empowerment, and affordability will be identified and coded. This will allow comparisons between community perspectives and the statistical data gathered earlier in the study. The analysis will provide a nuanced understanding of the social implications of the proposed Wi-Fi infrastructure and the lived experiences of community members.

4.7. Ethical Considerations

To ensure the ethical integrity of the research, all participants will be fully briefed on the purpose and scope of the study. Informed consent will be obtained in writing before participation, and all personal information will be kept confidential. The research will comply with ethical guidelines as set by the relevant academic institution, and approval will be obtained from the institutional review board.

4.8. Data Triangulation and Validity

Finally, the use of data triangulation cross-referencing qualitative and quantitative findings will help reinforce the trustworthiness and accuracy of the results, ensuring that both measurable trends and personal insights are reflected in the analysis. Triangulation increases the robustness of the conclusions drawn from the study, providing a clearer understanding of the complex issues at hand.

4.9. Conclusion

This mixed-methods research design aims to provide a comprehensive analysis of public Wi-Fi availability and the integration of smart streetlight infrastructure in Bloemfontein. By combining quantitative and qualitative approaches, the study will offer valuable insights into the potential for smart infrastructure to bridge the digital divide in urban South Africa, contributing to more reasonable access to digital resources.

CHAPTER 5: Prototype

The proposed solution addresses the digital divide in Bloemfontein by transforming existing municipal streetlight infrastructure into public internet access points. This is achieved by integrating Wi-Fi routers into streetlights, using their existing power supply to minimize additional infrastructure costs. The approach is both cost-effective and sustainable, enhancing digital access in underserved urban areas while supporting smart city development.

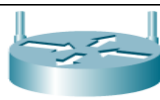
To effectively plan, visualize, and simulate the deployment, two key tools were used:

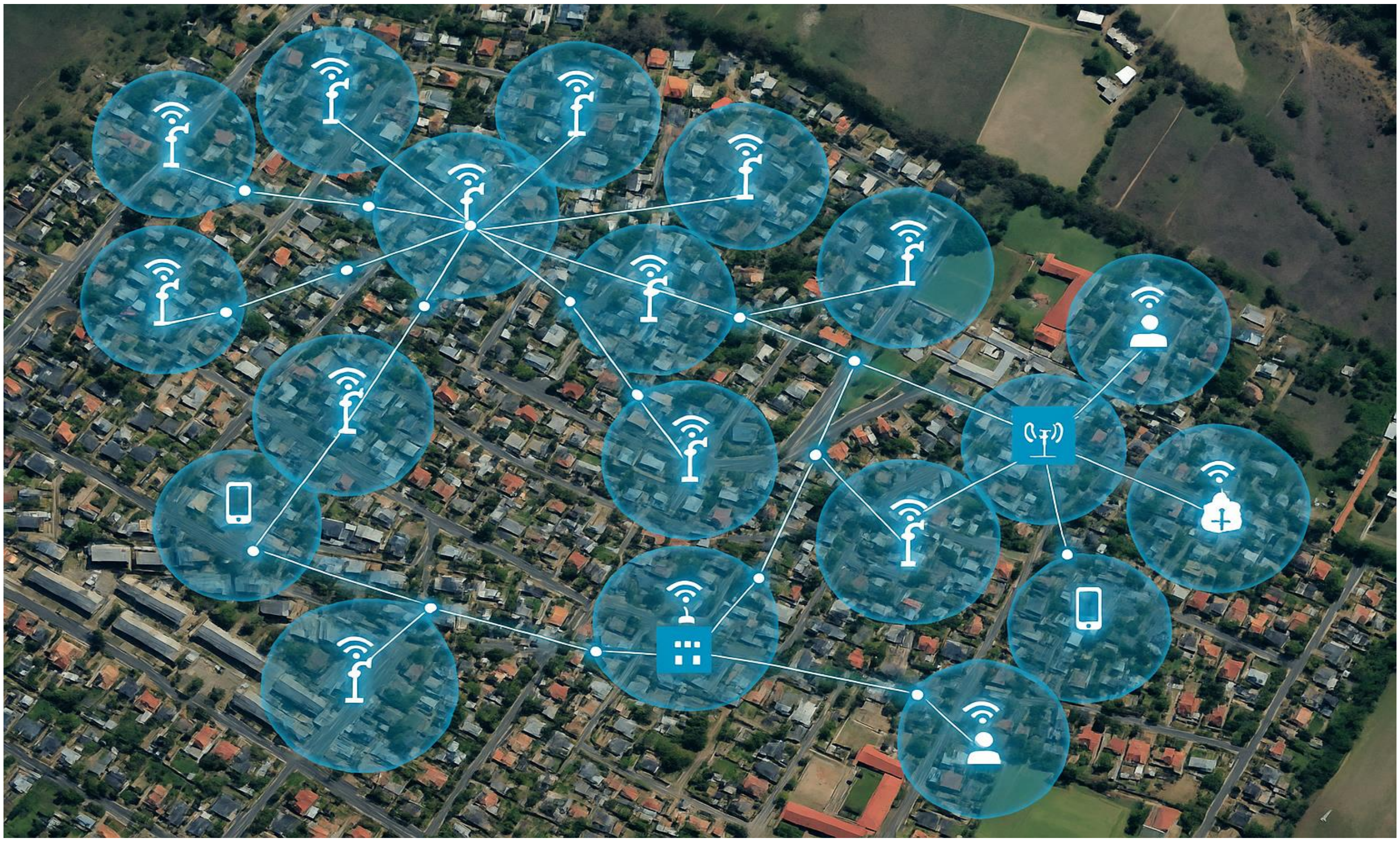
- **Figma** was utilized to design a visual prototype showing the physical placement of routers and the user interface for interacting with the system.
- **Cisco Packet Tracer** was used to simulate the **network architecture**, including IP configuration, routing, and connectivity between nodes.

5.1 Wi-Fi Network Coverage Simulation Cisco Packet Tracer:

The Cisco Packet Tracer simulation models the deployment of Wi-Fi routers on each streetlight within the Brandwag neighbourhood. The coverage zones of these routers are designed to overlap, ensuring uninterrupted internet access across the area. Each router is configured with DHCP and appropriate IP addressing, simulating a fully functional network that supports communication between devices and user connectivity.

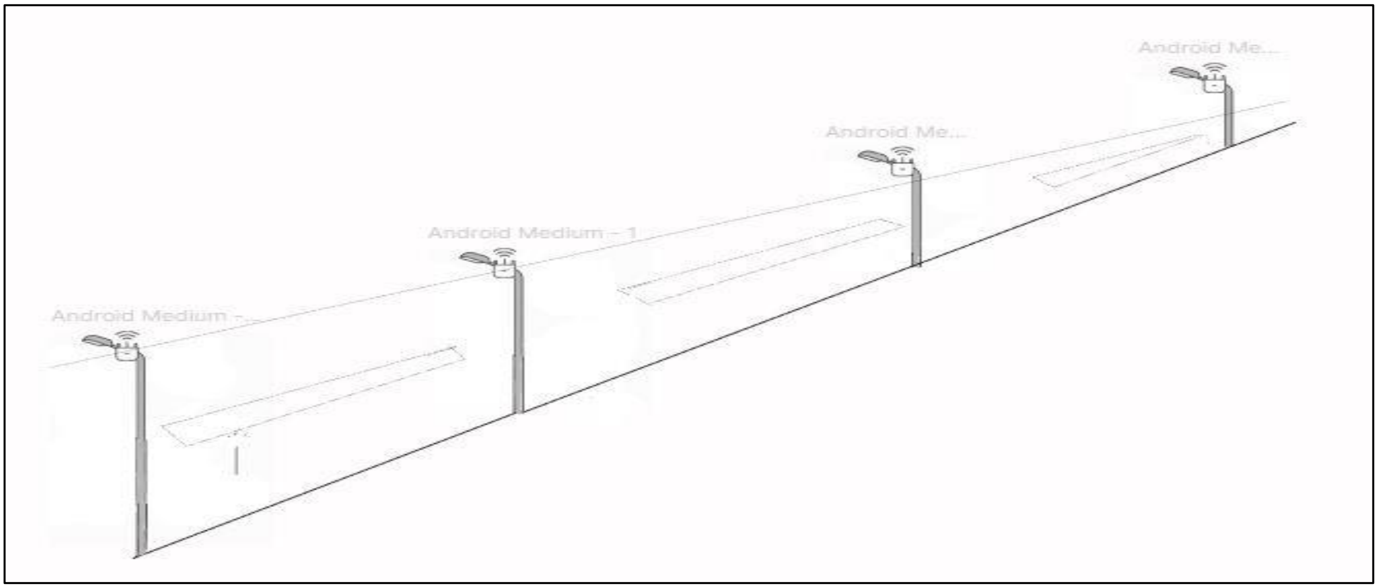
KEY

	Smart streetligts

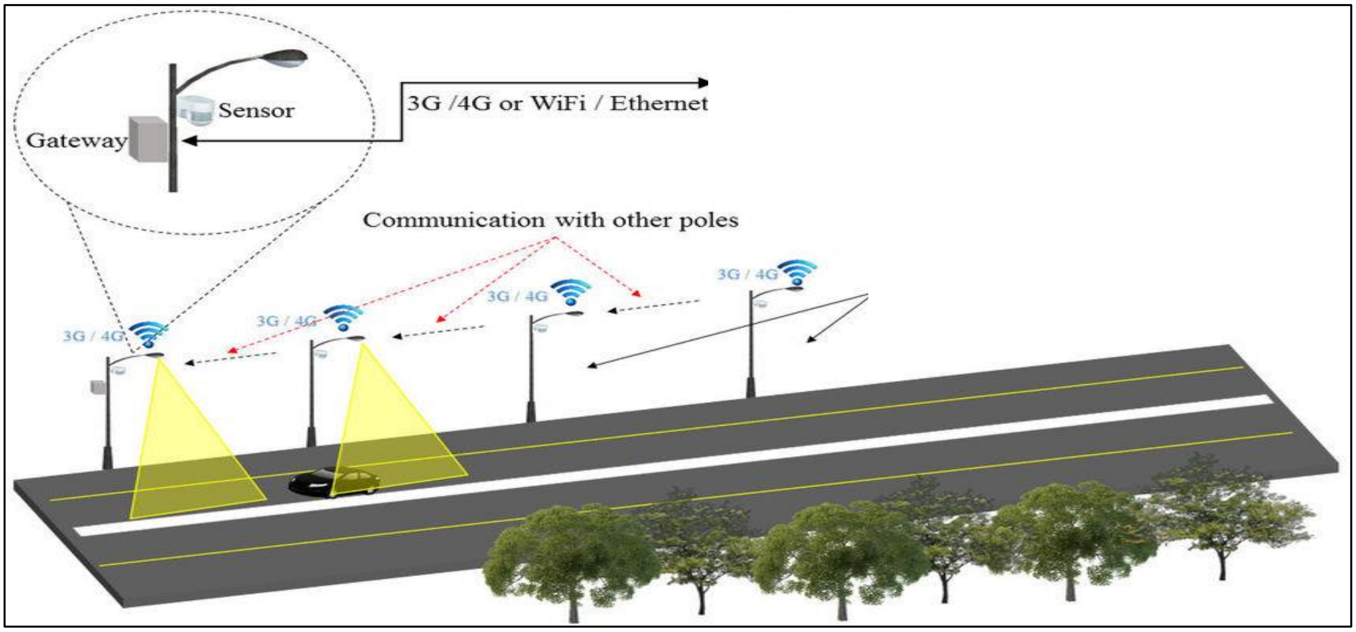


5.2 Secure Router Installation and Signal Optimization Figma Design :

The Figma prototype visualizes the elevated placement of Wi-Fi routers on top of each streetlight. This placement enhances signal range while reducing the risk of physical tampering. Strategic positioning also improves signal propagation and minimizes interference from common urban obstructions such as buildings and trees. This design supports critical system requirements, including security, performance, and ease of maintenance.

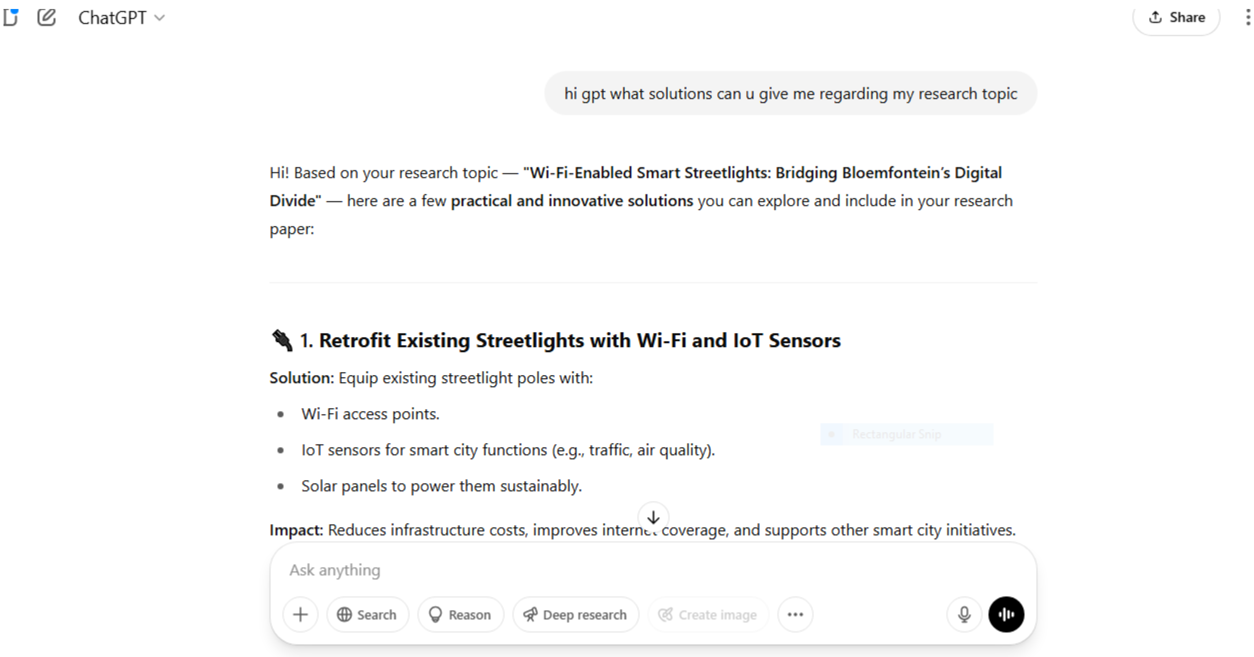


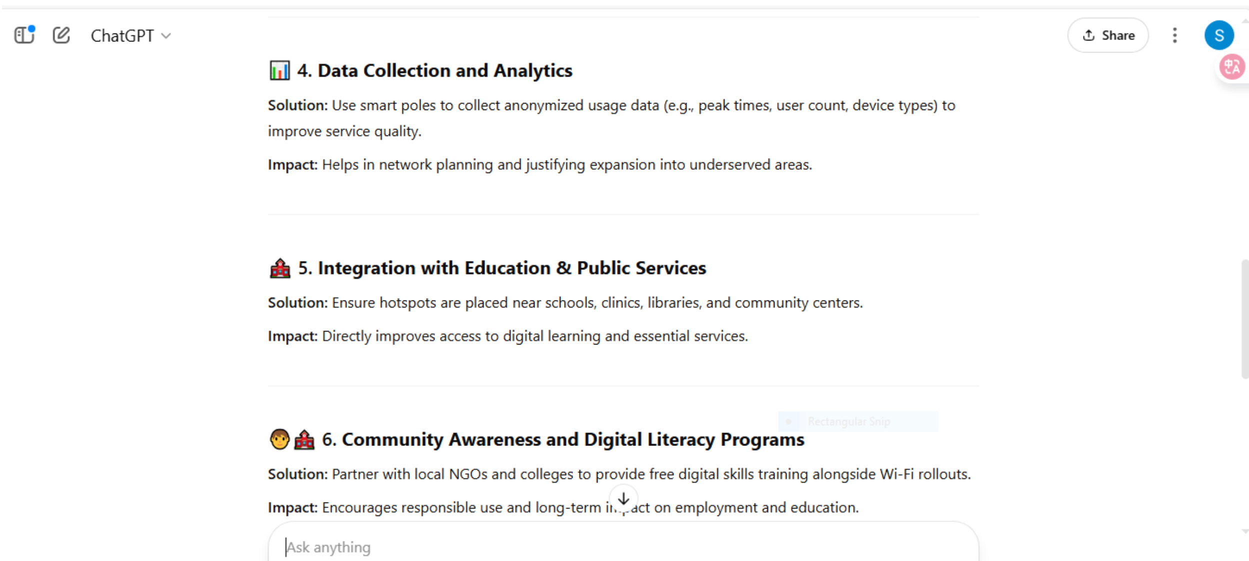
This image illustrates the conceptual design presented in the prototype, where routers are securely installed at the top of streetlights to extend Wi-Fi coverage across urban areas.



These visual prototypes highlight the seamless integration of Wi-Fi technology into street lighting systems, contributing to a smarter, more connected urban environment.

CHAPTER 6: APPLICATION AND CRITIQUE OF CHATGPT.





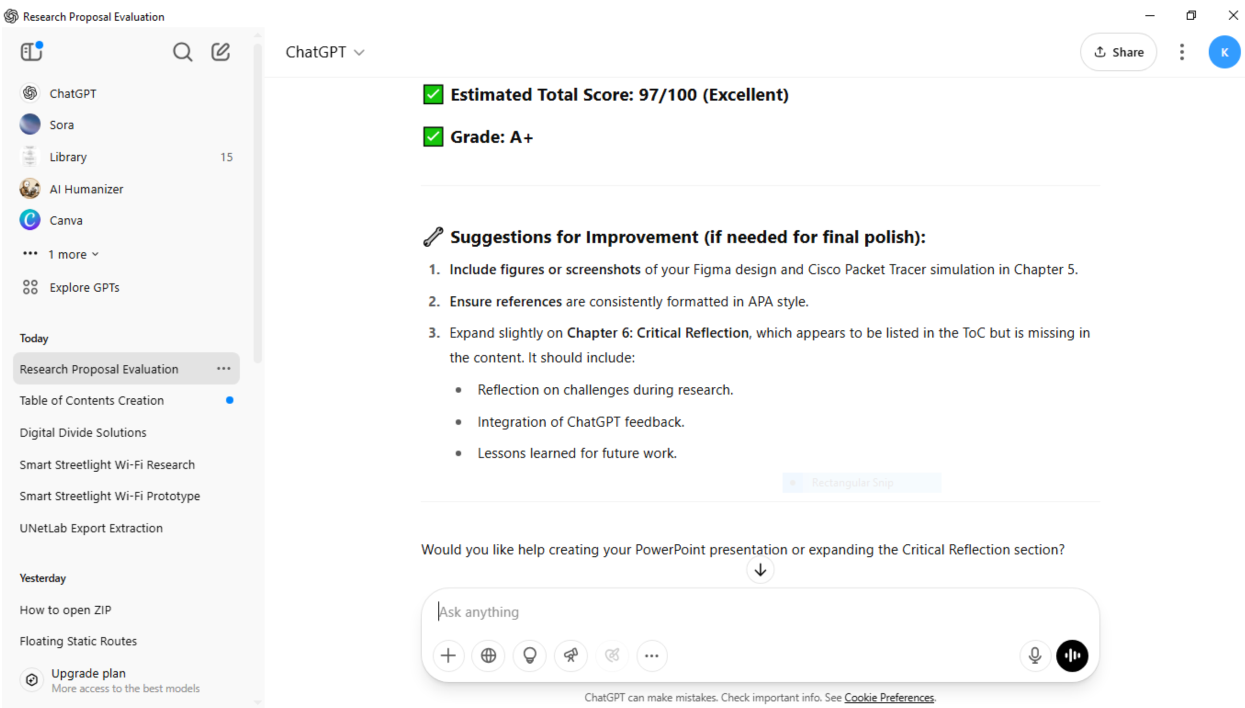
6.1 Critical reflection of the feedback from ChatGP

Using ChatGPT in my research has been both insightful and convenient. One of the main strengths is its ability to quickly provide well-structured explanations, summarize complex academic content, and assist in organizing my thoughts. It has helped me improve the clarity and flow of my writing, especially when drafting sections like the introduction, literature review, and methodology.

The tool is also useful for generating ideas, checking grammar, and ensuring my writing stays formal and academic.

Another strength is that ChatGPT can simulate academic-style responses and help me understand how to phrase certain arguments or concepts better. It acts like a brainstorming partner, especially when I am stuck or unsure how to approach a specific part of the research.

6.2 Critical reflection of the feedback from ChatGP.



I mostly agree with the evaluation ChatGPT gave me. The feedback was accurate, fair, and aligned with my own self-assessment. One area it suggested for improvement was adding figures to the prototype section, this was already done in my submission, but ChatGPT could not access or view the images within the uploaded document. Therefore, the critique regarding the lack of visuals does not fully apply.

The suggestion to critically synthesize the literature review was fair; although I had attempted to incorporate synthesis, I acknowledge that time constraints limited how deeply I could engage with comparative analysis. The strengths highlighted, such as my research methodology, clarity of the problem statement, and alignment between research questions and objectives were correctly identified.

References

Agramelal, F., Sadik, M., Moubarak, Y. and Abouzahir, S., 2023. *Smart Street Light Control: A Review on Methods, Innovations, and Extended Applications*. *Energies.*

Biundini, I.Z., Pinto, M.F., Honório, L.M., Capretz, M.A.M., Timotheo, A.O., Dantas, M.A.R. and Villela, P.C., 2024. LoRaCELL-Driven IoT Smart Lighting Systems: Sustainability in Urban Infrastructure. *Sensors*, 24(2).

Ditlhake matone, 2024. Digital divide deepens South Africa struggles with high broadband cost.

Donati Dante, 2023. Mobile Internet access and political outcomes Evidence from South Africa. *I exploit the staggered introduction of 3G mobile Internet technology across South Africa to estimate its impact on political participation, electoral competition*.

Dordley Lucinda, 2018. Streetlights to expand Cape Town's WiFi access. *the City of Cape Town has signed an agreement with mobile network and internet provider, Cell C to expand the public's access to WiFi across the city.*

Gehlot, A., Alshamrani, S.S., Singh, R., Rashid, M., Akram, S.V., Alghamdi, A.S. and Albogamy, F.R., 2021. Internet of things and long-range-based smart lampposts for illuminating smart cities. *Sustainability (Switzerland)*, 13(11).

Greedts Christopher, 2016. *DEVELOPING SMART PUBLIC WI-FI IN SOUTH AFRICA*. independent communications authority of South Africa, 2023. *The state of the ICT sector in South Africa 2023*.

Masia Laticia, 2024. South Africa Connect.

Mlaba Khanyi, 2021. How Is South Africa's Digital Divide Making Inequality Worse in the Country. *South Africa's digital divide can be broken down into three factors.*

Orejon-Sanchez, R.D., Andres-Diaz, J.R. and Gago-Calderon, A., 2021. Autonomous photovoltaic led urban street lighting: Technical, economic, and social viability analysis based on a case study. *Sustainability (Switzerland)*, 13(21).

Oza, M., Hinsu, L. and Goradiya, B.C., 2021. Intelligent and multipurpose smart pole. In: *Journal of Physics: Conference Series*. IOP Publishing Ltd. Pahlavan, K. and Krishnamurthy, P., 2021. *Evolution and Impact of Wi-Fi Technology and Applications: A Historical Perspective*. *International Journal of Wireless Information Networks.*

du Toit, B.R. and Stimie, J.E., 2023. TOWARDS SMART CITIES IN SOUTH AFRICA: EVOLUTION, DEFINITIONS, AND FUTURE CITIES. *South African Journal of Industrial Engineering*, 34(1), pp.85–96.